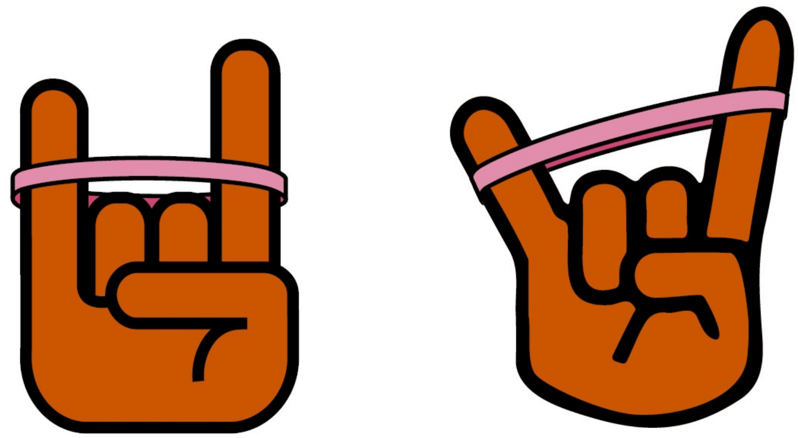
Lab 5

Hooke'em: Investigating Linear Elasticity



(Graphic by Savannah Hollan)

Physical Models: Hooke's Law

Analysis Tools: T-score, regime of validity, weighted averaging, propagation of uncertainty

Experimental Systems: Uniaxial tensile test

Equipment: Weight set, standing frame, clamps, carabiners, electronic scale on TA table, meter stick, calipers, micrometers

Safety Concerns: None

Introduction

In your prelab, you were introduced to Hooke's law. Hooke's law describes the linear relationship between an applied force and the resulting displacement for many materials:

$$|F| = k|x| \tag{5.1}$$

Where F is a force, x is the displacement from equilibrium, and k is a constant which captures the "stretchy-ness" of the material.

However, it's also the case that Hooke's Law isn't perfectly accurate for all forces, for all materials. A material is said to have "Hookean" or "linearly elastic" behavior if it obeys Hooke's law in a range of deformations that material would reasonably experience. For example, steel is considered to have linearly elastic behavior, despite the fact that it begins to yield and the behavior is no longer linear under extremely large deformations. In lab this week, your goal is to test the elastic behavior of an extensible material of your choosing. This is called a uni-axial tensile test and is common in materials science and engineering.

Part 1. Finding a Hookean Regime

You will need to apply a variety of forces and determine a range for which Hooke's law is accurate. You will also need to find the proportionality constant, k, for that linear regime.

Include plots of F vs. x and k vs. x in your lab notes. This will help you visualize the linearity of the material's response.

You may need the following partial derivatives if the method you choose requires propagation of uncertainty.

$$\frac{\partial k}{\partial F} = \frac{1}{x} \tag{5.2}$$

$$\frac{\partial k}{\partial x} = \frac{-F}{x^2} \tag{5.3}$$

$$\frac{\partial F}{\partial m} = g \tag{5.4}$$

Quick Check: Start by playing around with the hanging masses. Try adding a few to the bottom clamp and observe how the object deforms. Does the deformation seem linear or does the material stretch a lot and then plateau even as more weight is added? Or vice versa?

Designing Your Experiment: Now, write out a plan for a high-precision test of your hypothesis. Use the quick check to inform the range of masses you test in part one. Try to test over a range that includes a linear and nonlinear regime. Be sure to consider how you will analyze your data to determine which regimes are linear and nonlinear.

Check in with Another Group: Briefly discuss your data and analysis plans with another group. They may provide you with additional ideas and feedback. Note one similarity and one difference between your approaches or hypotheses.

Conducting Your Experiment: Go ahead and give it a shot. You can change your method around while you're at it, you don't need to treat your previous decisions as a contract. Just make sure to record any changes. Make sure all your data is labeled clearly with units, and that you keep track of uncertainty and try to minimize it. Keep working until you're confident in your conclusion.

Forming a Conclusion: What did you find? Was there a linear and nonlinear regime? Consider plotting F vs. x and k vs. x, like you did in the prelab, in order to form your conclusion. List a few improvements you could make and alternative ranges you could investigate in a next iteration. Additionally, consider what may be happening to the material that is producing the results you see.

Check in with Another Group: Check back in with the same group you discussed things with before. How do your results compare?

Part 2. Exploring Further

In part one, you probably got one of three results:

1. **Hooke's law applied over the entire range of forces that you tested.**

2. **Hooke's law applied to only a subset of the forces you tested.**

3. **Hooke's law did not apply over the entire range of forces you tested.**

If the first option is the case for your data...

- Try investigating a different range of forces, in search of a non-linear regime.
- OR Try investigating a different material altogether.
- OR Redesign your experiment to reduce the uncertainty of k.

If the second option is the case for your data...

- Try investigating the linear regime more closely.
- OR try investigating the non-linear regime more closely.
- OR Examine the transition between the linear and non-linear regime more closely.

If the third option is the case for your data...

- Try investigating a different range of forces.
- OR try investigating within a range of forces you already tested more closely, in search of a linear regime.
- OR try investigating a different material altogether.

Designing Your Experiment: Now, write out a plan for a high-precision test of your hypothesis. Use the results of part one to inform the range of masses you test in part two. Be sure to consider your analysis methods. Will they differ from what you did in part one?

Conducting Your Experiment: Go ahead and give it a shot. You can change your method around while you're at it, you don't need to treat your previous decisions as a contract. Just make sure to record any changes. Make sure all your data is labeled clearly with units, and that you keep track of uncertainty and try to minimize it. Keep working until you're confident in your conclusion.

Forming a Conclusion: What did you find? Was it what you expected, based on your results from part one? List a few improvements you could make and alternative variables you could investigate in a next iteration. Additionally, consider what may be happening to the material that is producing the results you see.

Check in with Another Group: Check back in with the same group you discussed things with before. How do your results compare?